

DATA STRUCTURE

LAB PROGRAM

14. Write a C program to implement the Tree Traversals (Inorder, Preorder, Postorder)

```
#include <stdio.h>

#include <stdlib.h>

struct Node {
    int data;
    struct Node *left, *right;
};

struct Node* createNode(int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->left = newNode->right = NULL;
    return newNode;
}

void inorder(struct Node* root) {
    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}
```

```
void preorder(struct Node* root) {  
    if (root != NULL) {  
        printf("%d ", root->data);  
        preorder(root->left);  
        preorder(root->right);  
    }  
}
```

```
void postorder(struct Node* root) {  
    if (root != NULL) {  
        postorder(root->left);  
        postorder(root->right);  
        printf("%d ", root->data);  
    }  
}
```

```
int main() {  
    struct Node* root = createNode(1);  
    root->left = createNode(2);  
    root->right = createNode(3);  
    root->left->left = createNode(4);  
    root->left->right = createNode(5);  
  
    printf("Inorder Traversal: ");  
    inorder(root);  
  
    printf("\nPreorder Traversal: ");
```

```
preorder(root);

printf("\nPostorder Traversal: ");
postorder(root);

return 0;
}
```

	Output
▲	Inorder Traversal: 4 2 5 1 3 Preorder Traversal: 1 2 4 5 3 Postorder Traversal: 4 5 2 3 1 === Code Execution Successful ===